

From Vectors to Rung 4

A ground-up build of the ER=EPR persistent homology project

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How to read this

This document is meant to be read straight through, top to bottom, with nothing skipped. The rule I've imposed on myself: **no term is used before it is defined**, and no result is asserted without either a proof or an explicit label saying it's imported from the literature.

Assumed background: linear algebra (vector spaces, bases, matrices, eigenvalues, inner products), calculus, and introductory physics (Newton's laws, energy, waves). Everything else is built here.

Three kinds of marked box appear throughout:

PROVEN. Derived here, from things already established. You should be able to reproduce it cold.

IMPORTED. True, standard, and used by us — but not derived here. These are dependencies of the project, and they belong in the paper's proven-vs-conjectured ledger.

CHOICE. A convention or modelling decision, not forced by anything. Worth knowing these exist so you never mistake one for a discovery.

That last category is the one that bit us in Rung 1, so I flag it aggressively.

Notation reference

Symbol	Meaning	Defined in
$\mathcal{H}$	Hilbert space	§3
$ \psi\rangle$	a state vector ("ket")	§3
$\langle\psi $	its conjugate transpose ("bra")	§3
$\langle\phi \psi\rangle$	inner product, a complex number	§3
$ \psi\rangle\langle\psi $	outer product, a matrix	§8
H	Hamiltonian (energy operator)	§5
$E_n, n\rangle$	energy eigenvalues and eigenvectors	§5

Symbol	Meaning	Defined in
$\otimes$	tensor product	§6
ρ	density matrix	§8
Tr, Tr_B	trace, partial trace	§8
S	von Neumann entropy	§9
β	inverse temperature, $1/k_B T$	§10
Z	partition function	§10
τ	Euclidean (imaginary) time	§11
ψ_i	Majorana fermion operator	§17
N	number of Majorana fermions	§17
J_{ijkl}	random SYK couplings	§18
$G(\tau)$	two-point function	§19
C	Schwarzian coupling constant	§20
φ	the natural transformation we're after	§27

PART I — QUANTUM MECHANICS FROM SCRATCH

§1. What a "state" is

Start classically, where your intuition already works.

A ball flies through the air. To predict its entire future you need its position $\vec{r}$ and its momentum $\vec{p}$. Six numbers. Feed them to $\vec{F} = m\vec{a}$ and the trajectory is fixed forever, forward and backward in time.

Those six numbers are the **state**. The definition that generalizes:

A **state** is a complete specification of a system at one instant — complete in the sense that it is everything the dynamical law needs in order to produce the next instant.

Two properties of the classical state. Quantum mechanics keeps one and breaks the other.

Property 1 — the state is a sufficient input to the dynamics. You never need history. Give me $(\vec{r}, \vec{p})$ now and I don't care how the ball got there. Quantum mechanics keeps this exactly: the Schrödinger equation is first order in time, and the state at t determines the state at $t + dt$.

Property 2 — the state assigns a definite value to every observable. Classically, given $(\vec{r}, \vec{p})$, the kinetic energy is $p^2/2m$, the angular momentum is $\vec{r} \times \vec{p}$. Every observable is a function on the state space, so all of them have values simultaneously.

Quantum mechanics breaks Property 2. A quantum state is still *complete* — there is no hidden extra layer, nothing is being withheld — but for most observables it determines only a **probability distribution** over outcomes.

A **quantum state** is a complete specification of a system such that, for every observable, it determines the probability distribution of measurement outcomes.

The word "complete" is doing heavy lifting. The randomness is not ignorance. We will build, in §7, a state of two particles that is completely known and yet gives a perfect coin flip when you measure either particle. That is not a gap in the description. It is a fact about the world.

One structural difference to notice now, because it's what the next section changes. The classical state space of a ball is $\mathbb{R}^6$ — a *set of points*. There's no useful notion of adding two states. What is "ball here moving left" plus "ball there moving right"? Nothing. The question is meaningless.

Quantum states can be added. That's the next section, and it's the whole reason the mathematics is linear algebra.

§2. Superposition

Superposition is an experimental fact before it is a piece of mathematics. So: the fact first, then the formalization.

2.1 The experiment

Fire electrons at a barrier with two slits in it. Behind the barrier, a detector screen. Send them one at a time — so slowly that there is only ever one electron in the apparatus.

Each electron arrives at a single spot. A dot on the screen. Electrons are particles; they land in one place. Fine.

Now three runs:

Slit 2 closed. Electrons pass through slit 1. Accumulate many dots, get a distribution $P_1(x)$: a blob behind slit 1.

Slit 1 closed. Get $P_2(x)$: a blob behind slit 2.

Both slits open. You expect $P_1(x) + P_2(x)$. Each electron went through *some* slit, so the counts should add. Two blobs.

You do not get two blobs. You get **interference fringes** — bands where electrons pile up, alternating with bands where *no electron ever lands*, including locations that were perfectly well populated when only one slit was open.

Read that last clause twice. **Opening a second slit made certain outcomes impossible.** Adding a route to the destination removed ways of arriving.

There is no story in which each electron went through one definite slit that survives this. In such a story the counts must add, and they don't. And it isn't electrons colliding — there's only ever one in the machine. Each electron interferes with itself.

2.2 The formalization

Probabilities don't add. But the fringe pattern looks exactly like the water-wave and light-wave patterns you've seen, and for waves, *amplitudes* add and then you take the intensity.

So try this: assign to each path a complex number, an **amplitude**. Amplitudes add. Probability is the modulus squared of the total:

$$\psi_{\text{tot}} = \psi_1 + \psi_2, \quad P = |\psi_1 + \psi_2|^2.$$

Expand:

$$|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + \underbrace{2 \operatorname{Re}(\overline{\psi_1} \psi_2)}_{\text{interference term}} = P_1 + P_2 + \text{interference}.$$

Write the amplitudes in polar form, $\psi_1 = r_1 e^{i\theta_1}$ and $\psi_2 = r_2 e^{i\theta_2}$. Then the interference term is

$$2r_1 r_2 \cos(\theta_2 - \theta_1).$$

It depends only on the **relative phase** $\theta_2 - \theta_1$, and it can be negative. When $\theta_2 - \theta_1 = \pi$ and $r_1 = r_2$, the total probability is exactly zero — a dark fringe. An outcome each path allowed alone, and both paths together forbid.

That reproduces the experiment. And it forces three things:

States must be addable. $\psi_1 + \psi_2$ has to be a legitimate state, or you cannot describe an electron with both slits open.

Amplitudes must be complex. Real numbers give you a sign, $\pm$, but the fringes vary *smoothly* from constructive to destructive as you move across the screen. That needs a continuously varying phase $e^{i\theta}$. Complex numbers are forced, not a computational convenience.

Only relative phase is physical. Multiply both ψ_1 and ψ_2 by the same $e^{i\gamma}$ and P is unchanged. So an overall phase on the whole state is invisible. Keep this — it comes back in §8 as a structural fact.

2.3 The definition

Superposition. If $|\psi_1\rangle$ and $|\psi_2\rangle$ are physically realizable states, then for any complex α, β not both zero, the combination $\alpha|\psi_1\rangle + \beta|\psi_2\rangle$ (normalized) is also a physically realizable state — genuinely distinct from both, and distinct from any probabilistic mixture of the two.

2.4 A superposition is not a mixture

This is where intuition usually breaks, so let's kill the confusion immediately with a concrete calculation. Write the two basis states $|0\rangle$ and $|1\rangle$ (formalized in §4; for now just two distinguishable alternatives). Compare:

(A) the superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

(B) a coin flip: the system is in $|0\rangle$ with probability $\frac{1}{2}$, in $|1\rangle$ with probability $\frac{1}{2}$, and I simply don't know which

Measure in the $\{|0\rangle, |1\rangle\}$ basis. **Identical statistics:** 50/50 either way. Indistinguishable.

Now measure in a different basis. Define

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

These are orthogonal and normalized, so they're a perfectly good basis to measure in.

(A) is $|+\rangle$. So you get outcome $+$ with probability $|\langle + | + \rangle|^2 = 1$. **Certainty.**

(B): if the system is $|0\rangle$, then $|\langle + | 0 \rangle|^2 = \frac{1}{2}$. If it's $|1\rangle$, also $\frac{1}{2}$. Total: 50/50. **Coin flip.**

Different physics, experimentally distinguishable, no ambiguity.

The distinction to carry forward. A superposition is a *definite state* — you know everything about it, and asking the right question gets a certain answer. A mixture is *ignorance* — no question has a certain answer. §8 gives the formalism that separates them mechanically.

So "the cat is alive and dead at the same time" is a bad slogan. Better: **the cat is in a state for which "alive?" is the wrong question.** It has a definite answer to some other question — one we have no everyday word for, because cats don't do this.

§3. Hilbert space

Now we can say what the state space is, and why.

3.1 The vector space axioms, and which ones physics gives us

A complex vector space is a set V with addition $V \times V \rightarrow V$ and scalar multiplication $\mathbb{C} \times V \rightarrow V$, satisfying eight axioms. Check them against §2:

Axiom	Physical source
Closure under addition	Superposition principle, directly (§2.3)
Closure under complex scaling	Amplitudes are complex numbers you can scale
Commutativity $\psi_1 + \psi_2 = \psi_2 + \psi_1$	"Both slits open" has no ordering
Associativity	Three slits: grouping them doesn't change the pattern
Distributivity	Rescaling the beam rescales each path's amplitude equally
Zero vector, additive inverses	Not physical — see below

The honest caveat: the zero vector is not a state (a state must be normalizable, and $\|0\| = 0$), and $|\psi\rangle, 2|\psi\rangle, e^{i\theta}|\psi\rangle$ all describe the *same* physical state. So:

CHOICE. Physical states are the nonzero vectors of a complex vector space, modulo nonzero scalar multiplication — a *projective* space. We work in the ambient linear space anyway, because linear algebra is a finished subject with spectral theorems in it, and $V \setminus \{0\}$ modulo scaling is an awkward place to do algebra.

So the vector space structure isn't a definition handed down from on high, and it isn't a full derivation either. It's the minimal structure that makes interference expressible, tidied into a standard object.

3.2 Bra-ket notation

Write a state as $|\psi\rangle$, Dirac's "ket." It's a vector; picture a column:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The "bra" $\langle\psi|$ is its conjugate transpose, a row vector:

$$\langle\psi| = (\bar{\alpha} \quad \bar{\beta}).$$

Then $\langle\phi|\psi\rangle$ is a row times a column — a $1 \times n$ times an $n \times 1$ — giving a single complex number. That's the inner product, and the notation is engineered so it looks like a bracket split in half. That is the entire joke.

3.3 The inner product, and where it comes from

An inner product is a map $\mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ with:

Conjugate symmetry: $\langle \phi | \psi \rangle = \overline{\langle \psi | \phi \rangle}$

Linearity in the second slot: $\langle \phi | (a\psi_1 + b\psi_2) \rangle = a\langle \phi | \psi_1 \rangle + b\langle \phi | \psi_2 \rangle$

Positive definiteness: $\langle \psi | \psi \rangle \geq 0$, with equality only for the zero vector

CHOICE. Physicists put linearity in the second slot; mathematicians usually put it in the first. Watch this when reading across fields.

Where does the inner product come from physically? From probability. The **Born rule** (§4) says the probability of finding a system prepared in $|\psi\rangle$ to be in state $|\phi\rangle$ is $|\langle \phi | \psi \rangle|^2$. Then:

positive definiteness $\Leftrightarrow$ probabilities are non-negative

conjugate symmetry $\Leftrightarrow$ "find ψ in ϕ " and "find ϕ in ψ " are equally likely

normalization $\langle \psi | \psi \rangle = 1 \Leftrightarrow$ probabilities sum to one

The inner product *is* the probability structure, written in linear algebra.

3.4 The definition

A **Hilbert space** $\mathcal{H}$ is a complex vector space with an inner product, which is complete in the induced norm $\|\psi\| = \sqrt{\langle \psi | \psi \rangle}$.

Completeness means every Cauchy sequence converges to a limit inside the space. This is an analysis condition, and:

In finite dimensions it is automatic. Every finite-dimensional complex inner product space is complete. So for everything in this project, "Hilbert space" means " $\mathbb{C}^n$ with the standard inner product."

Completeness only earns its keep in infinite dimensions (wavefunctions on the real line, quantum field theory). File it and move on. Every Hilbert space we touch — one qubit, two qubits, SYK with $N = 18$ — is finite dimensional.

§4. One qubit: observables and measurement

The smallest interesting case. Take $\mathcal{H} = \mathbb{C}^2$, fix an orthonormal basis, name it:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

4.1 Observables are Hermitian operators

An **observable** is a Hermitian operator $A = A^\dagger$ ($\dagger$ = conjugate transpose).

Why Hermitian? Because of the **spectral theorem**: a Hermitian operator on a finite-dimensional complex space has all-real eigenvalues and an orthonormal basis of eigenvectors. Measurement outcomes are real numbers, and we need a basis to expand in. Hermiticity delivers both.

The standard qubit observables are the Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Check σ_z : eigenvector $|0\rangle$ with eigenvalue $+1$, eigenvector $|1\rangle$ with eigenvalue -1 . Real eigenvalues, orthonormal eigenvectors. ✓

4.2 The Born rule

The Born rule (axiom). Measure observable A , with eigenvectors $|a_k\rangle$ and eigenvalues a_k , on a system in state $|\psi\rangle$. You obtain outcome a_k with probability $|\langle a_k|\psi\rangle|^2$, and the state collapses to $|a_k\rangle$.

This is an axiom. It is not derived from the rest of the formalism, and attempts to derive it are a live research area. Take it as given.

The expectation value over many repetitions is

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

PROVEN. Expand $|\psi\rangle = \sum_k c_k |a_k\rangle$ with $c_k = \langle a_k | \psi \rangle$. Then $\langle \psi | A | \psi \rangle = \sum_{k,l} \bar{c}_l c_k a_k \langle a_l | a_k \rangle = \sum_k |c_k|^2 a_k$, which is exactly $\sum_k (\text{probability of } a_k) \times a_k$. ■

Why states are unit vectors. Probabilities must sum to 1:

$$\sum_k |\langle a_k | \psi \rangle|^2 = \langle \psi | \psi \rangle$$

by completeness of the orthonormal eigenbasis. So $\langle \psi | \psi \rangle = 1$ is exactly the normalization of probability.

Global phase drops out, relative phase doesn't. Replace $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$: every $|\langle a_k | \psi \rangle|^2$ is unchanged. But $|+\rangle$ and $|-\rangle$ — which differ only by an internal relative sign — are *orthogonal*, hence perfectly distinguishable. This is §2.2 point 3, now a theorem in the formalism.

§5. Time evolution, the Hamiltonian, and the eigenbasis

5.1 The Hamiltonian

The **Hamiltonian** H is the observable corresponding to energy. It is Hermitian, and it generates time evolution via the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle.$$

For time-independent H this integrates immediately:

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle,$$

where the exponential of an operator is defined by its power series. Note the equation is **first order in t** — the state now determines the state later, with no reference to history. That's Property 1 from §1, preserved exactly.

CHOICE. From here on I set $\hbar = 1$, standard practice. It just fixes units so that energy and inverse-time are the same thing.

5.2 Specifying a system

We can now say precisely what a "system" is, because it's a term I'll use constantly:

A **system** is a Hilbert space $\mathcal{H}$ together with a Hamiltonian H on it. $\mathcal{H}$ says what states are available; H says how they evolve and what the energies are.

That's the whole definition. "One qubit" is $\mathcal{H} = \mathbb{C}^2$ with some H . "The SYK model" (§18) is a $2^{N/2}$ -dimensional space with a specific H . Note that a system is a *modelling decision* — you decide which degrees of freedom you're describing. Nature doesn't draw the box.

5.3 The eigenbasis, and what's stored in the energies

By the spectral theorem there is an orthonormal basis $\{|n\rangle\}$ with

$$H|n\rangle = E_n|n\rangle,$$

all E_n real. This is the **energy eigenbasis**. In it, H is the diagonal matrix $\text{diag}(E_0, E_1, \dots)$.

Two facts make this *the* basis to work in.

(a) The E_n are the possible outcomes of an energy measurement. By the Born rule, measuring H returns some eigenvalue and never anything else. The spectrum is the complete menu of energies the system can

have.

(b) Energy eigenstates are stationary.

PROVEN. $e^{-iHt}|n\rangle = e^{-iE_nt}|n\rangle$, since $|n\rangle$ is an eigenvector of H and hence of any power series in H . That's a global phase, which §4.2 showed is unphysical. So nothing about an energy eigenstate ever changes. ■

Consequently, expanding any state in the eigenbasis makes time evolution trivial:

$$|\psi(0)\rangle = \sum_n c_n |n\rangle \implies |\psi(t)\rangle = \sum_n c_n e^{-iE_nt} |n\rangle.$$

Diagonalizing H is solving the system. Every dynamical question becomes easy afterward, and essentially impossible before.

This is also where the computational wall in our project lives. For SYK with N Majorana fermions, H is a $2^{N/2} \times 2^{N/2}$ matrix. At $N = 18$ that's 512×512 — instant. At $N = 40$ it's about a million by a million — no. That's why "accessible N " in Rung 4 means $N \leq 18$, and it is a hard arithmetic fact, not a lack of effort.

§6. Two systems: the tensor product

Now the step that generates everything interesting.

Axiom. Given systems A and B with Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, the composite system has Hilbert space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$.

Tensor product, not direct sum. This is the single most consequential structural choice in quantum mechanics, so let's make it concrete.

If $\{|i\rangle_A\}$ is a basis of $\mathcal{H}_A$ (dimension m) and $\{|j\rangle_B\}$ a basis of $\mathcal{H}_B$ (dimension n), then $\{|i\rangle_A \otimes |j\rangle_B\}$ is a basis of $\mathcal{H}_{AB}$, which therefore has dimension mn .

Dimensions multiply. They do not add.

Two qubits: $2 \times 2 = 4$, with basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ where $|ij\rangle$ abbreviates $|i\rangle_A \otimes |j\rangle_B$. A general two-qubit state:

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle, \quad \sum |c_{ij}|^2 = 1.$$

Classically, two bits take 2 numbers to describe. Quantum mechanically, 4 complex amplitudes. For n qubits: 2^n . **That exponential is why quantum systems are hard to simulate**, and it is exactly the wall from §5.3.

Concretely, $\otimes$ on vectors is the Kronecker product:

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}.$$

Ordering: the entries run through B 's index fastest, matching the basis ordering 00, 01, 10, 11.

Subsystems. A and B are called **subsystems** of the composite. When I later say "trace out B ," B is whichever tensor factor we're not looking at:

Setting	A	B
Two qubits shared by two people	Alice's qubit	Bob's qubit
Thermofield double (§12)	left copy L	right copy R
Black hole	outside the horizon	inside the horizon
Any lab experiment	the apparatus	the rest of the universe

CHOICE. The factorization $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ is a decision. The same vector can be more or less entangled depending on how you split. In this project the split is always physically forced (two copies of a system; or left boundary vs. right boundary), so the ambiguity never bites us — but it exists, and it's worth knowing.

§7. Entanglement

Definition. A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is **product** (or separable) if there exist $|\phi\rangle_A$ and $|\chi\rangle_B$ with $|\psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B$. Otherwise it is **entangled**.

That's the entire definition. No mysticism.

7.1 The Bell state, with proof

Define

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

PROVEN. $|\Phi^+\rangle$ is entangled.

Suppose it factors. Then for some $a, b, c, d \in \mathbb{C}$,

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

Matching coefficients against $|\Phi^+\rangle$:

$$ac = \frac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0, \quad bd = \frac{1}{\sqrt{2}}.$$

From $ad = 0$: either $a = 0$ or $d = 0$. If $a = 0$ then $ac = 0 \neq \frac{1}{\sqrt{2}}$. If $d = 0$ then $bd = 0 \neq \frac{1}{\sqrt{2}}$. Contradiction either way. ■

Do this once by hand. Afterwards "entanglement" stops being a word and becomes a statement about the rank of the coefficient matrix $C = (c_{ij})$ — which, as §9 shows, is exactly what it is.

7.2 Why it's strange

$|\Phi^+\rangle$ is a state of **maximal knowledge** about the pair. It's a single vector; nothing is unspecified.

But measure qubit A alone in the $\{|0\rangle, |1\rangle\}$ basis. By the Born rule you get 0 with probability $\frac{1}{2}$ and 1 with probability $\frac{1}{2}$. Maximal ignorance about the part.

Classically this is impossible. Complete knowledge of a whole implies complete knowledge of its pieces. Quantum mechanically, information can be stored **entirely in the correlations**, with none of it in the individual subsystems.

That sentence is the seed of this entire project. Sit with it before continuing.

To say it precisely we need a way to describe "the state of A alone" — which cannot be a vector, since no vector in $\mathbb{C}^2$ reproduces the statistics of half a Bell pair. That's the next section.

§8. Density matrices and the partial trace

8.1 Outer products

$|\psi\rangle$ is a column, $\langle\psi|$ is a row. Column times row is a **matrix** (order matters: row times column was the inner product, a number).

$$|\psi\rangle\langle\psi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \bar{\alpha} & \bar{\beta} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\bar{\beta} \\ \bar{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

Read the entries:

diagonal: $|\alpha|^2, |\beta|^2$ — the probabilities of measuring 0 or 1.

off-diagonal: the **coherences**, carrying the relative phase. These are what distinguish a superposition from a mixture.

8.2 The density matrix

Definition. The **density matrix** of a state $|\psi\rangle$ is $\rho = |\psi\rangle\langle\psi|$. More generally, a density matrix is any operator

that is Hermitian, positive semidefinite, and has $\text{Tr } \rho = 1$.

Expectation values become $\langle A \rangle = \text{Tr}(\rho A)$.

PROVEN. For $\rho = |\psi\rangle\langle\psi|$: $\text{Tr}(|\psi\rangle\langle\psi|A) = \sum_k \langle k|\psi\rangle\langle\psi|A|k\rangle = \langle\psi|A(\sum_k |k\rangle\langle k|)|\psi\rangle = \langle\psi|A|\psi\rangle$, using cyclicity of the trace and $\sum_k |k\rangle\langle k| = I$. ■

Pure vs. mixed. ρ is **pure** iff $\rho^2 = \rho$ iff $\text{Tr}(\rho^2) = 1$; it then equals $|\psi\rangle\langle\psi|$ for some vector. Otherwise it is **mixed**, and represents a genuine probabilistic ensemble.

Return to §2.4:

(A) superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$: $\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, and $\rho_A^2 = \rho_A$. Pure.

(B) coin flip: $\rho_B = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I/2$, and $\rho_B^2 = I/4 \neq \rho_B$. Mixed.

Same diagonal, different off-diagonal. The formalism separates superposition from mixture mechanically, exactly as promised.

Also note: $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ give the *identical* density matrix, since the phases cancel between ket and bra. The unphysical redundancy of §3.1 is quotiented out automatically.

8.3 The partial trace

Definition. The **partial trace** over B is the unique linear map $\rho_{AB} \mapsto \rho_A$ satisfying

$$\text{Tr}_A[\rho_A O_A] = \text{Tr}_{AB}[\rho_{AB} (O_A \otimes I_B)] \quad \text{for every observable } O_A.$$

That definition is the physics. Read it: ρ_A **is defined to be whatever object reproduces every measurement A could possibly make without touching B .**

Explicitly, in components: with $|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B$,

$$\rho_{AB} = \sum_{ijkl} c_{ij} \bar{c}_{kl} |i\rangle\langle k|_A \otimes |j\rangle\langle l|_B,$$

and tracing the B slot (which sends $|j\rangle\langle l| \mapsto \delta_{jl}$) while leaving the A slot alone:

$$\rho_A = \sum_{ik} \left(\sum_j c_{ij} \bar{c}_{kj} \right) |i\rangle\langle k|_A, \quad \text{i.e.} \quad \boxed{\rho_A = C C^\dagger}$$

where C is the matrix of coefficients c_{ij} . Clean and computable.

This is the quantum version of marginalizing a joint probability distribution: given $P(x, y)$, you get $P(x) = \sum_y P(x, y)$. The partial trace is that, upgraded to handle coherences.

8.4 Operational content

Here's the payoff, and the reason ρ_A is the central object rather than a lossy convenience.

PROVEN. Every measurement performable on A alone depends on the global state only through ρ_A . An observer with access only to A can measure exactly the observables of the form $O_A \otimes I_B$ — do something to A , nothing to B . By the defining property in §8.3, $\langle O_A \otimes I_B \rangle = \text{Tr}_A[\rho_A O_A]$. ■

Two different global states with the same ρ_A are **indistinguishable to that observer forever**, no matter how clever the experiment. ρ_A is a sufficient statistic for everything local.

This is what "operational content" means: content cashed out in experiments and outcomes rather than in formalism. The operational content of a state, for a local observer, is exactly its reduced density matrix — no more, no less.

This will matter enormously in §12, where we find a *pure* state whose reduced density matrix is *exactly thermal*. By the theorem above, one-sided physics is then exactly thermal physics — not approximately, not by analogy. Operationally identical, forever.

8.5 Worked example: the Bell state

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ has coefficient matrix $C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. So

$$\rho_A = CC^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

Maximally mixed. $\text{Tr}(\rho_A^2) = \frac{1}{2}$, the smallest possible value for a qubit, while ρ_{AB} is pure with $\text{Tr}(\rho_{AB}^2) = 1$.

Where did the coherence go? $|\Phi^+\rangle$'s off-diagonal terms were coherences between $|00\rangle$ and $|11\rangle$ — correlations *between* the two systems. Discard B and they have nowhere to live.

They weren't destroyed. They were never stored in A to begin with. Tracing out B doesn't lose A 's information; it reveals that A never had any.

§9. Entropy and the Schmidt decomposition

9.1 Von Neumann entropy

Definition. $S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_k \lambda_k \log \lambda_k$, summed over the eigenvalues λ_k of ρ (with $0 \log 0 := 0$).

This is the Shannon entropy of the eigenvalue distribution. It measures how mixed ρ is:

ρ pure $\Rightarrow$ eigenvalues $(1, 0, \dots, 0) \Rightarrow S = 0$.

$\rho = I/d$ maximally mixed $\Rightarrow$ eigenvalues all $1/d \Rightarrow S = \log d$, the maximum.

When ρ_A comes from tracing a pure global state, $S(\rho_A)$ is called the **entanglement entropy** of the split.

Product state: ρ_A is pure, $S_A = 0$.

Bell state: $\rho_A = I/2$, $S_A = \log 2$ — one bit, maximal for a qubit.

$S_A = 0$ if and only if the global state is a product state. **Entanglement entropy detects entanglement and grades it.** This is the number that will eventually be identified with the area of a black hole horizon (§15).

9.2 Schmidt decomposition

****Theorem (Schmidt).**** Any pure $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ can be written

$$|\psi\rangle = \sum_{k=1}^r \lambda_k |k\rangle_A \otimes |k\rangle_B$$

with $\lambda_k > 0$, $\sum_k \lambda_k^2 = 1$, and $\{|k\rangle_A\}, \{|k\rangle_B\}$ orthonormal sets in their respective spaces.

PROVEN. Write $|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B$ and take the singular value decomposition $C = U \Sigma V^\dagger$. Absorb U into a new basis for A and V into a new basis for B ; the singular values are the λ_k . ■

So the **Schmidt decomposition is the SVD wearing a physics hat**. The number of nonzero terms r is the **Schmidt rank**, and

$$r = 1 \iff \text{product state.}$$

Look back at §7.1: we were proving that $\mathrm{rank}(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}) \neq 1$. The Bell proof was a rank computation in disguise.

9.3 Two consequences we will use constantly

(a) Both sides have the same spectrum. From the Schmidt form, $\rho_A = \sum_k \lambda_k^2 |k\rangle\langle k|_A$ and $\rho_B = \sum_k \lambda_k^2 |k\rangle\langle k|_B$. Same eigenvalues, hence

$$S_A = S_B \quad \text{always,}$$

no matter how differently sized A and B are. Entanglement is symmetric.

(b) The Schmidt coefficients are a free design parameter. Any set of positive numbers summing (in square) to 1 gives a legitimate pure state, and it fully determines ρ_A 's spectrum.

That second point is the door to Part II. Look at the Schmidt form and ask:

What if I choose $\lambda_n = e^{-\beta E_n/2} / \sqrt{Z}$, with $|n\rangle$ the energy eigenstates of a Hamiltonian H ?

Then ρ_A has eigenvalues $e^{-\beta E_n} / Z$. We're about to see (§10) that this is precisely the thermal distribution — meaning we will have built a **pure entangled state of two copies whose restriction to one copy is exactly a thermal state**.

That object is the thermofield double, and it is the central character of this project.

PART II — TEMPERATURE, EUCLIDEAN TIME, AND THE THERMOFIELD DOUBLE

§10. What "thermal" means, precisely

10.1 The Gibbs state

Definition. A system with Hamiltonian H is in a **thermal state** at inverse temperature β if its density matrix is

$$\rho_{\text{th}}(\beta) = \frac{e^{-\beta H}}{Z(\beta)}, \quad Z(\beta) = \text{Tr } e^{-\beta H}, \quad \beta = \frac{1}{k_B T}.$$

Z is the **partition function**; it exists only to make $\text{Tr } \rho = 1$.

In the energy eigenbasis this is diagonal with entries

$$p_n = \frac{e^{-\beta E_n}}{Z},$$

the Boltzmann distribution from intro statistical mechanics. High-energy states are exponentially suppressed. Limits:

$\beta \rightarrow 0$ (infinitely hot): all p_n equal. Maximally mixed, entropy maximal.

$\beta \rightarrow \infty$ (absolute zero): only the ground state survives. Pure, entropy zero.

CHOICE. I set $k_B = 1$ from here, so $\beta = 1/T$ exactly. Again just units.

10.2 Why *this* form and not some other

The Gibbs state isn't arbitrary. It is the unique state that maximizes von Neumann entropy subject to a fixed average energy.

PROVEN (sketch). Maximize $S = -\text{Tr} \rho \log \rho$ subject to $\text{Tr} \rho = 1$ and $\text{Tr}(\rho H) = \bar{E}$. Introduce Lagrange multipliers μ and β for the two constraints, vary with respect to ρ :

$$-\log \rho - 1 - \mu - \beta H = 0 \implies \rho \propto e^{-\beta H}.$$

Normalizing fixes the prefactor to $1/Z$. ■

So a thermal state is the **maximally noncommittal state consistent with knowing only the average energy**, and β is literally the Lagrange multiplier conjugate to energy. That's what temperature *is*, structurally. Worth doing this variation yourself once; it takes an afternoon and it demystifies temperature permanently.

10.3 What "looks thermal" means in this project

When I say a subsystem "looks thermal," I mean something exact, not a vibe:

ρ_A is equal to $e^{-\beta H_A} / Z$ for some β .

Not approximately. Not "behaves similarly to." Equal, as matrices.

And by §8.4, if ρ_A is exactly thermal then **every experiment the A -observer can ever do returns thermal statistics**. There is no measurement that reveals otherwise, because ρ_A is a sufficient statistic for all of them.

10.4 The conceptual crack

Normally, a thermal state signals **ignorance**. Your coffee is at 350 K because it exchanges energy with the room and you don't track every molecule. Thermality = missing information about a larger system.

Section 12 will produce a thermal ρ_A arising from a **pure** global state — no missing information at all.

Thermality is not always ignorance. Sometimes it is entanglement. And by §8.4, the local observer cannot tell the difference, ever.

That is the conceptual hinge of the entire subject. Black holes have a temperature and an entropy (§15). The thermofield double says those can come from entanglement with a partner system — and the geometry joining them is a wormhole.

§11. Euclidean time: why temperature is a circle

This section is short and it is the most important piece of technology in the document. Everything about the filtration in Part V rests on it.

Real-time evolution is e^{-iHt} (§5.1). Substitute $t = -i\tau$ for real τ :

$$e^{-iHt} \longrightarrow e^{-H\tau}.$$

The variable τ is called **Euclidean time** (the name comes from the fact that this rotation turns the Minkowski metric $-dt^2 + dx^2$ into the Euclidean $d\tau^2 + dx^2$).

11.2 The identification

Now stare at the thermal density matrix:

$$\rho_{\text{th}} = \frac{e^{-\beta H}}{Z}.$$

PROVEN (by inspection). $e^{-\beta H}$ is exactly the Euclidean time evolution operator evaluated at $\tau = \beta$. **The thermal density matrix is time evolution by an imaginary amount, and the amount is the inverse temperature.**

11.3 Why it's a circle

$Z = \text{Tr } e^{-\beta H}$ is a *trace*. A trace identifies the input index with the output index — it glues the beginning to the end.

So Euclidean time, in a thermal system, is **periodic with period β** . It lives on a circle of circumference β .

Concretely, any thermal correlation function of an operator obeys

$$G(\tau + \beta) = \pm G(\tau)$$

(+ for bosons, − for fermions — the sign comes from anticommutation, §17). Either way, the domain is $\tau \in [0, \beta)$ with the ends identified.

This is the single most important structural fact for our project. Thermal physics and Euclidean-time-on-a-circle are the same subject. It's why temperature and geometry can talk to each other at all. And it is the reason your entire filtration is built on **the thermal circle**: the circle isn't a modelling convenience we chose, it's what a thermal system's Euclidean time *is*.

Two immediate corollaries we'll cash in later:

Turning β up or down changes the *circumference of a circle*. Temperature is a geometric parameter.

Points on that circle are the natural place to put a point cloud, and distances between them can be built from correlators. Part V.

§12. The thermofield double

12.1 Construction

Take a system with Hamiltonian H , energy eigenbasis $\{|n\rangle\}$, eigenvalues E_n (§5.3). Make **two identical copies**, called L and R , so the composite Hilbert space is $\mathcal{H}_L \otimes \mathcal{H}_R$ (§6). Define

$$|\text{TFD}(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_n e^{-\beta E_n/2} |n\rangle_L \otimes |n\rangle_R$$

Note this is exactly the Schmidt form of §9.2 with the coefficients chosen as $\lambda_n = e^{-\beta E_n/2} / \sqrt{Z}$.

Normalization checks out: $\sum_n \lambda_n^2 = \frac{1}{Z} \sum_n e^{-\beta E_n} = 1$ by definition of Z .

So the TFD is not an exotic invention. **It is the generic pure state of a doubled system, with one specific choice of Schmidt coefficients.** The only creative act is choosing $e^{-\beta E_n/2}$.

12.2 The theorem

PROVEN. $\text{Tr}_R |\text{TFD}(\beta)\rangle \langle \text{TFD}(\beta)| = \frac{e^{-\beta H}}{Z} = \rho_{\text{th}}(\beta)$.

By §9.3(a), the reduced state of a Schmidt-decomposed pure state is $\rho_L = \sum_n \lambda_n^2 |n\rangle \langle n|_L$. Substituting $\lambda_n^2 = e^{-\beta E_n} / Z$:

$$\rho_L = \sum_n \frac{e^{-\beta E_n}}{Z} |n\rangle \langle n|_L = \frac{e^{-\beta H}}{Z}.$$

The last equality is just the statement that $e^{-\beta H}$ is diagonal in the energy eigenbasis with entries $e^{-\beta E_n}$. ■

The TFD is a pure state of the pair whose restriction to either copy is exactly thermal. Everything promised in §10.4 is now established, in three lines.

By §8.4, an observer confined to L experiences a system at temperature $1/\beta$, with no experiment able to reveal the purity of the global state. The thermality is entirely due to entanglement with R .

12.3 Temperature is the entanglement knob

Compute the entanglement entropy using §9.1:

$$S_L = - \sum_n \frac{e^{-\beta E_n}}{Z} \log \frac{e^{-\beta E_n}}{Z} = \beta \langle E \rangle + \log Z,$$

which is the ordinary thermodynamic entropy. Limits:

β	Physical regime	ρ_L	S_L	TFD structure
$\rightarrow 0$	infinitely hot	maximally mixed	maximal	maximally entangled
finite temperature	$1/\beta$ thermal		intermediate	partially entangled
$\rightarrow \infty$	absolute zero	ground state, pure 0		product state $ 0\rangle_L 0\rangle_R$

Check the last row directly: as $\beta \rightarrow \infty$, the factor $e^{-\beta E_n/2}$ kills every term except the lowest energy, leaving a single product term. Schmidt rank 1. Unentangled, by §9.2.

Turning the temperature dial is turning the entanglement dial. One real parameter β , monotonically interpolating between maximal entanglement and none.

That is the entire content of Rung 2 (§29), and it's why Rung 2 was a β -sweep.

12.4 Euclidean picture

Combining with §11: the TFD has a geometric description. $Z = \text{Tr } e^{-\beta H}$ is a Euclidean path integral over a circle of circumference β . Cutting that circle open at one point gives a path integral over an *interval* of length β — with two open ends, which are the two boundaries L and R .

The TFD is the state prepared by a Euclidean path integral over half the thermal circle, with the two cut ends becoming the two copies.

Hold this picture. In §16 the same cut circle becomes a spatial slice through a wormhole, and the two ends become the two asymptotic boundaries of an eternal black hole.

PART III — THE GRAVITY SIDE

Minimal build. I'm importing more here than in Part I, and I'll say exactly what.

§13. The bare minimum of general relativity

Spacetime is a manifold with a metric. The metric $g_{\mu\nu}$ is the object that assigns lengths to curves and angles between them. Given a path, its length is $\int \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} d\lambda$. That's all "geometry" means here: a rule for measuring.

Curvature is built from the metric. The Ricci scalar R is a number at each point measuring how curved the space is there, computed from second derivatives of g . $R = 0$ is flat, $R < 0$ is hyperbolic (saddle-like), $R >$

0 is spherical.

Geodesics are curves of extremal length. In flat space, straight lines. On a sphere, great circles. **Geodesic length between two boundary points** will be our distance function on the gravity side, so this is the one geometric quantity that actually enters the pipeline.

Anti-de Sitter space (AdS) is the maximally symmetric spacetime with constant *negative* curvature. Two properties matter:

It has a **boundary at infinite distance but finite time** — light can reach it and come back in finite time. So AdS behaves like a box, and it makes sense to say a quantum system "lives on the boundary."

Its symmetry group is the same as the conformal group of one dimension lower. That coincidence is the seed of holography (§15).

AdS₂ is the two-dimensional case: one space, one time. That's the arena for JT gravity (§21). It has $R = -2$ in units where the AdS radius is 1.

Horizon. A surface beyond which signals cannot return to a distant observer. The defining feature of a black hole.

§14. Black hole thermodynamics

Three imported results. All are standard and none are derived here.

****IMPORTED (Bekenstein–Hawking).**** A black hole has entropy proportional to the *area* of its horizon:

$$S_{\text{BH}} = \frac{A}{4G}.$$

IMPORTED (Hawking). A black hole radiates as a thermal body at a temperature set by its surface gravity, $T_H = \kappa/2\pi$.

IMPORTED (chaos bound, Maldacena–Shenker–Stanford). In any thermal quantum system, the Lyapunov exponent governing the growth of certain out-of-time-order correlators obeys $\lambda_L \leq 2\pi/\beta$. Black holes **saturate** this bound.

Why these matter to us:

Entropy is an area. Entropy is an information-theoretic quantity (§9.1); area is geometric. The formula equates them. That's the first hint that information and geometry are the same subject.

Temperature exists, so §10 and §11 apply to black holes: a black hole has a β , hence a thermal circle.

Saturating the chaos bound is a signature. Any quantum system that saturates it is behaving like a black hole. In §19 we'll meet one that does.

§15. Holography, and the two statements we actually use

IMPORTED (AdS/CFT, Maldacena). Certain quantum theories without gravity, living on the boundary of an AdS spacetime, are *equivalent* to gravitational theories in the interior ("bulk"). Same physics, two descriptions, related by a dictionary.

"Equivalent" is strong: the Hilbert spaces are isomorphic and the dynamics match. Everything happening in a $(d + 1)$ -dimensional bulk with gravity is encoded in a d -dimensional boundary theory without it. Hence "holography."

The one dictionary entry we use:

IMPORTED (Ryu-Takayanagi). The entanglement entropy of a boundary region A equals the area of the minimal bulk surface anchored to ∂A , divided by $4G$:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G}.$$

RT is the sharp version of "entanglement is geometry." An information-theoretic quantity on the left; a geometric one on the right.

FLAG — read this twice. RT is exactly the relation that made Rungs 1 and 2 tautological. If you *use* RT to compute the gravity-side quantity, then agreement between the two sides is guaranteed by construction, because you put it in by hand. See §32. The whole reason Rung 3 onward uses persistent homology is to get a comparison that RT does not automatically satisfy.

§16. ER=EPR

Two 1935 papers, both by Einstein and collaborators, published two months apart:

EPR (Einstein-Podolsky-Rosen): the paper that identified entanglement, intending it as a paradox.

ER (Einstein-Rosen): the paper that found the "bridge" solution — two asymptotic regions of spacetime joined through a throat. A wormhole.

For decades these were unrelated. Then:

IMPORTED (Maldacena, "Eternal black holes in AdS"). The thermofield double state of a holographic boundary theory is dual to the **eternal AdS black hole**: a spacetime with *two* asymptotic boundaries, joined by a non-traversable Einstein-Rosen bridge.

Line up the pictures. From §12.4, the TFD is prepared by a Euclidean path integral over half the thermal circle, with two cut ends. On the gravity side, that same computation prepares a spatial slice of the eternal black hole geometry — and the two cut ends are the two asymptotic boundaries.

Same object. Two languages:

Quantum side	Gravity side
Pure state on $\mathcal{H}_L \otimes \mathcal{H}_R$	Two-boundary spacetime
ρ_L thermal at $1/\beta$ (§12.2)	Black hole at Hawking temperature $1/\beta$
Entanglement entropy S_L (§12.3)	Horizon area over $4G$ (§14)
$\beta \rightarrow 0$: maximal entanglement	Long, wide wormhole
$\beta \rightarrow \infty$: product state	Bridge pinched off; two disconnected spacetimes

Read the last two rows as one sentence:

ER=EPR (Maldacena-Susskind). Entanglement *is* the connectivity of spacetime. Remove the entanglement and the geometric connection disappears; add it and the two regions become joined.

This is the conjecture the project is about. And now we can be exact about what's wrong with it as stated.

"Entanglement is geometry" has no falsification condition. What would you measure? What would a disagreement look like? If entanglement went up and the wormhole didn't get longer — longer by which yardstick, measured how? A claim you can't lose isn't a claim about the world. Part V is the fix, and §25 is where the fix is stated precisely.

PART IV — SYK AND JT GRAVITY

We need a concrete pair: a specific quantum system and a specific gravity theory, so that "compute both sides" is an actual computation.

§17. Fermions and Majorana notation

17.1 Anticommutators

You know the commutator $[A, B] = AB - BA$. The **anticommutator** is

$$\{A, B\} = AB + BA.$$

Bosons are built with commutators, fermions with anticommutators. That sign difference is where the Pauli exclusion principle comes from.

17.2 What a Majorana fermion is

An ordinary fermion has a creation operator $c^\dagger$ and a *distinct* annihilation operator c — particle and antiparticle differ.

A **Majorana fermion** is its own antiparticle. Its operator is Hermitian:

$$\psi_i^\dagger = \psi_i.$$

You obtain them by splitting an ordinary fermion into "real and imaginary parts":

$$\psi_1 = \frac{1}{\sqrt{2}}(c + c^\dagger), \quad \psi_2 = \frac{i}{\sqrt{2}}(c^\dagger - c).$$

Both are manifestly Hermitian. So:

A Majorana is half of an ordinary fermion. N Majoranas correspond to $N/2$ ordinary fermions, giving a Hilbert space of dimension $2^{N/2}$. (N must be even.)

17.3 Reading $\{\psi_i, \psi_j\} = \delta_{ij}$

This compact line contains two statements.

$i = j$: $\{\psi_i, \psi_i\} = 2\psi_i^2 = 1$, so $\psi_i^2 = \frac{1}{2}$. Each Majorana squares to a constant.

$i \neq j$: $\psi_i\psi_j = -\psi_j\psi_i$. Distinct Majoranas anticommute; swapping two costs a minus sign.

CHOICE. The normalization is a convention. Many papers use $\{\psi_i, \psi_j\} = 2\delta_{ij}$, giving $\psi_i^2 = 1$. Ours gives $\psi_i^2 = \frac{1}{2}$. Factors of 2 in downstream formulas depend on this, so it must be stated and held fixed. (This is exactly the class of thing that turns into an unexplained prefactor six weeks later.)

This algebra is a **Clifford algebra** — the same structure as Dirac gamma matrices.

17.4 Concretely, as matrices

$N = 2$ gives dimension $2^{2/2} = 2$, a single qubit:

$$\psi_1 = \frac{1}{\sqrt{2}}\sigma_x, \quad \psi_2 = \frac{1}{\sqrt{2}}\sigma_y.$$

Verify: both Hermitian ✓; $\psi_1^2 = \frac{1}{2}\sigma_x^2 = \frac{1}{2}I$ ✓; and $\{\sigma_x, \sigma_y\} = \sigma_x\sigma_y + \sigma_y\sigma_x = i\sigma_z - i\sigma_z = 0$ ✓.

For larger N , the **Jordan-Wigner construction** strings Pauli matrices together:

$$\psi_1 \sim \sigma_x \otimes I \otimes \cdots, \quad \psi_2 \sim \sigma_y \otimes I \otimes \cdots, \quad \psi_3 \sim \sigma_z \otimes \sigma_x \otimes \cdots$$

The trailing σ_z factors are what enforce anticommutation across different sites. It's finite, explicit, and boring — and it's exactly what the code does to build the $2^{N/2} \times 2^{N/2}$ matrices.

§18. The SYK model

18.1 The Hamiltonian

N Majorana fermions with all-to-all random four-body coupling:

$$H = \frac{1}{4!} \sum_{i,j,k,l=1}^N J_{ijkl} \psi_i \psi_j \psi_k \psi_l$$

with J_{ijkl} drawn from a Gaussian distribution with mean zero and variance

$$\langle J_{ijkl}^2 \rangle = \frac{3! J^2}{N^3}.$$

Term by term:

**** $\sum_{ijkl}$ **** runs over all ordered quadruples. Because the ψ 's anticommute (§17.3), J_{ijkl} is taken totally antisymmetric, so each *unordered* set $\{i, j, k, l\}$ is counted $4! = 24$ times — hence the $1/4!$.

Four-body: every four fermions interact simultaneously. Not two-body, not nearest-neighbour.

All-to-all, no geometry: the index i is a *label*, not a position. There is no lattice, no notion of "nearby," no spatial structure whatsoever. This is deliberate — geometry is supposed to *emerge*, not be inserted.

Random couplings: you draw a realization of J , compute, and average observables over many draws ("disorder averaging").

N^{-3} **scaling** is fixed by requiring the energy to be extensive (to scale like N). Count the number of terms and the scaling and it falls out. It is not a free knob.

FLAG. "Index i is a label, not a position" is the fact that generated the Rung 3 structural mismatch (§30). Remember it.

18.2 Why anyone cares

Three properties, and they're the reason SYK is a field rather than a curiosity:

Solvable at large N (§19).

Maximally chaotic — it saturates the chaos bound of §14.

Nonzero entropy at zero temperature.

Black holes are also all three. That coincidence is the entire motivation.

§19. What the large- N solution actually says

This is where, in earlier drafts, I asserted a chain of results you had no way to check. Here is the honest version: what each object *is*, and what status the claim has.

19.1 The two-point function

$$G(\tau) = \langle T \psi_i(\tau) \psi_i(0) \rangle$$

Reading it: perturb fermion i at Euclidean time 0, ask how correlated it is with itself at Euclidean time τ ; T denotes time-ordering; the angle brackets are a thermal expectation value (§10) *and* an average over disorder realizations (§18.1). By §11.3, τ lives on a circle of circumference β and $G(\tau + \beta) = -G(\tau)$.

This is the object the entire pipeline consumes. In Part V, $G(\tau_a - \tau_b)$ is what defines the distance between two points on the thermal circle, which defines the filtration, which defines the barcode. Everything downstream is a functional of G . If you understand nothing else in this section, understand that G is the input.

19.2 Schwinger-Dyson equations

A pair of coupled self-consistent equations for G and the **self-energy** Σ (which encodes the accumulated effect of interactions):

$$G(i\omega) = \frac{1}{-i\omega - \Sigma(i\omega)}, \quad \Sigma(\tau) = J^2 G(\tau)^3.$$

They're self-consistent in the same way mean-field theory is: the field determines the response determines the field. You solve them by iterating numerically until convergence. Concrete algorithm, no mystery.

19.3 Melonic diagrams

Perturbation theory generates infinitely many Feynman diagrams. At large N each diagram carries a power of $1/N$, and almost all of them vanish as $N \rightarrow \infty$. What survives is one specific nested family — diagrams that look like melon slices stacked inside melon slices.

Why it matters, concretely: ****because only melons survive, the self-energy equation is exactly $\Sigma = J^2 G^3$, with no other terms.**** The equations *close* — finitely many unknowns, no infinite hierarchy. That is what "solvable at large N " means.

19.4 The conformal IR

At strong coupling ($\beta J \gg 1$) and long times, drop the $-i\omega$ term in the SD equations (it's subleading), and the remaining equations admit a power-law solution

$$G(\tau) \propto \left[\frac{\pi}{\beta J \sin(\pi\tau/\beta)} \right]^{2\Delta}, \quad \Delta = \frac{1}{4}.$$

Status: derivable, moderate work. Plug a power-law ansatz into the reduced SD equations and match exponents; $\Delta = 1/4$ is forced. This is one of the few large- N results you could genuinely reproduce yourself in a sitting, and it's worth doing, because $\sin(\pi\tau/\beta)$ is the functional form that shows up everywhere downstream — including as the shape your Rung 4 fits were testing against.

Note the $\sin(\pi\tau/\beta)$: manifestly a function on the thermal circle, symmetric about $\tau = \beta/2$, singular as $\tau \rightarrow 0$. That singularity is the short-distance (UV) behaviour, and it becomes important in §31.

19.5 Reparametrization modes and the Schwarzian

The conformal solution has a large symmetry: it is invariant under reparametrizations $\tau \rightarrow f(\tau)$ of the thermal circle. But this symmetry is *both* spontaneously broken (a solution has to pick one f) and explicitly broken (the term we dropped in §19.4 isn't exactly zero).

When a symmetry is broken like that, the associated modes become light but not massless, and they dominate the low-energy physics. Here those modes are the reparametrizations $f(\tau)$, and their effective action is the **Schwarzian**:

$$S[f] = -C \int_0^\beta d\tau \text{Sch}(f, \tau), \quad \text{Sch}(f, \tau) = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2.$$

IMPORTED (Maldacena-Stanford). The derivation of the Schwarzian action from SYK's broken reparametrization modes, and the value of the coupling C in terms of N and J , are taken from the literature.

The constant C is the **Schwarzian coupling**. In the large- N dictionary it scales like N/J .

FLAG. C is the quantity your Rung 4 was fitting (§31). Its large- N value is *imported calibration*, not something the project measured — which is exactly how it is declared in `rung4_results.txt`, and correctly so.

§20. JT gravity

Jackiw-Teitelboim gravity is two-dimensional dilaton gravity:

$$I = -\frac{1}{2} \int d^2x \sqrt{g} \phi (R + 2) - \int_{\partial} \sqrt{h} \phi_b (K - 1).$$

Here g is the metric, R its Ricci scalar (§13), ϕ is a scalar field called the **dilaton**, and the second term is a boundary term (K is the extrinsic curvature of the boundary curve).

The key move: vary the action with respect to ϕ . Since ϕ appears linearly, its equation of motion is simply

$$R + 2 = 0 \implies R = -2.$$

So the bulk is *rigidly* AdS_2 . Not approximately, not on average — exactly, everywhere, as a constraint. There is no bulk graviton, no local gravitational degree of freedom, nothing propagating in the interior.

Where did the dynamics go? To the boundary. All that remains is **the shape of the boundary curve as it wiggles inside a fixed AdS_2** , parametrized by a function $f(\tau)$.

IMPORTED. The action for that boundary wiggle is the Schwarzian, $S[f] = -C \int \text{Sch}(f, \tau) d\tau$.

§21. The bridge: what SYK and JT actually share

Compare §19.5 and §20:

The low-energy sector of SYK is governed by a Schwarzian action for a reparametrization mode $f(\tau)$ on the thermal circle.

The *entire* dynamics of JT gravity is a Schwarzian action for a boundary reparametrization $f(\tau)$.

Same effective theory. That's the connection, and it is why SYK is called a model of near- AdS_2 holography.

Two honest caveats, both of which matter for the paper.

This is **not a full duality** in the AdS/CFT sense of §15. It is a shared universal low-energy sector. SYK has plenty of physics (all the UV, all the non-Schwarzian modes) that JT gravity does not describe at all.

Everything in this chain — melonic dominance, the Schwarzian action from SYK, the Schwarzian from JT, the value of C — is **imported**. We have not independently verified any of it. That's fine, you can't rederive all of physics. But it belongs in the ledger alongside the Maldacena-Stanford calibration, and it should be treated consistently.

The pairing we will actually test:

SYK thermofield double at inverse temperature β Eternal AdS_2 black hole

Two-point function $G(\tau)$ on the thermal circle Geodesic lengths between boundary points

Now we need something to compute on both sides.

PART V — TOPOLOGY, AND WHY IT'S THE RIGHT TOOL

§22. What we need and why nothing so far provides it

We have two descriptions of the same physical situation and want to test whether they agree. To do that we need a quantity $\mathcal{I}$ that we can compute on each side and then compare.

Three requirements, and they're not negotiable:

(a) Computable on both sides, into the same target. $\mathcal{I}$ eats an SYK correlation structure and outputs an object; $\mathcal{I}$ eats a JT geometry and outputs an object *of the same kind*. Then "equal or not" is a well-posed question rather than a metaphor.

(b) Invariant under the arbitrary choices made on each side. SYK doesn't care which basis you diagonalized in or how you labelled the fermions. JT doesn't care which bulk coordinates you chose. If $\mathcal{I}$ changed when you relabelled, then comparing the two sides would be comparing conventions, not physics.

(c) Computed independently on the two sides. This is the one that matters most and the one we got wrong first. Two agreeing computations are evidence **only if a disagreement was possible**.

The test to apply every single time: what input would have made these disagree? If the answer is "none," you have a tautology, not a result.

Requirement (b) is what the word **invariant** means, so let's be precise about it.

§23. Invariants

Definition. An **invariant** is a quantity assigned to an object that does not change under some specified class of transformations. Because it doesn't change, it is a property of the *object* rather than of your description of it.

Examples you already know:

Dimension of a vector space. Change basis however you like; dimension is fixed. So it's a fact about the space, not about coordinates.

Euler characteristic $\chi = V - E + F$ of a surface. Deform the surface however you like, retriangulate it, stretch it; χ doesn't move. Sphere: $\chi = 2$. Torus: $\chi = 0$.

Determinant, invariant under similarity transformations $A \mapsto PAP^{-1}$.

The pattern: an invariant converts "these look different / these look the same" into **a computation with a definite answer**. $\chi(\text{sphere}) = 2 \neq 0 = \chi(\text{torus})$, therefore no continuous deformation takes one to the other. That's a theorem, not an impression.

And the failure mode to watch: invariants can *agree* on genuinely different objects. χ can't tell a point from a disc (both 1). So a matching invariant is evidence, not proof — while a *mismatching* invariant is proof of difference. Asymmetric, and worth remembering when reading Rung 4.

§24. Homology, built up

We need an invariant that detects **holes**. Here's the construction, in stages.

24.1 Simplices and simplicial complexes

A **k -simplex** is the convex hull of $k + 1$ points in general position:

0-simplex: a point

1-simplex: an edge

2-simplex: a filled triangle

3-simplex: a solid tetrahedron

A **simplicial complex** is a collection of simplices that fit together properly: every face of a simplex in the collection is also in the collection, and any two simplices meet in a common face or not at all. Think of it as a combinatorial object — a list of vertices, a list of edges, a list of triangles, and so on — that happens to have a geometric realization.

The crucial distinction, which the entire subject rests on: **a hollow triangle (three vertices, three edges, no 2-simplex) is a different complex from a filled triangle (three vertices, three edges, one 2-simplex)**. The first has a hole; the second doesn't. Homology is the machinery that turns that intuition into a number.

Work with coefficients in a field (take $\mathbb{R}$, or $\mathbb{Z}/2$ for simplicity).

A **k -chain** is a formal linear combination of k -simplices. The set of k -chains is a vector space C_k with the k -simplices as basis.

The **boundary operator** $\partial_k : C_k \rightarrow C_{k-1}$ sends a simplex to the (signed) sum of its faces:

$$\partial_1(\text{edge } [a, b]) = [b] - [a]$$

$$\partial_2(\text{triangle } [a, b, c]) = [b, c] - [a, c] + [a, b]$$

PROVEN. $\partial_{k-1} \circ \partial_k = 0$: **the boundary of a boundary is zero.** Check on the triangle: $\partial_1 \partial_2[a, b, c] = ([c] - [b]) - ([c] - [a]) + ([b] - [a]) = 0$. The general case is the same cancellation, term by term. ■

Two subspaces of C_k :

Cycles: $Z_k = \ker \partial_k$ — chains with no boundary. A loop of edges is a 1-cycle: it closes up.

Boundaries: $B_k = \text{im } \partial_{k+1}$ — chains that are the boundary of something one dimension up. The rim of a filled triangle is a 1-boundary.

Because $\partial\partial = 0$, every boundary is a cycle: $B_k \subseteq Z_k$.

24.3 Homology

Definition. The **k -th homology group** is the quotient

$$H_k = Z_k / B_k = \frac{\ker \partial_k}{\text{im } \partial_{k+1}}.$$

In words: **cycles that are not boundaries.** Loops that close up but don't bound anything — i.e. loops with a hole inside them.

Interpretation by degree:

H_0 : connected components. Its dimension is the number of pieces.

H_1 : one-dimensional holes — loops. Circle: $\dim H_1 = 1$. Filled disc: $\dim H_1 = 0$.

H_2 : voids, enclosed cavities. Hollow sphere: $\dim H_2 = 1$.

The dimension of H_k is called the k -th **Betti number**, b_k .

Worked: **hollow triangle.** The loop $[a, b] + [b, c] + [c, a]$ is a cycle (its boundary telescopes to zero). Is it a boundary? $C_2 = 0$ — there are no 2-simplices — so $B_1 = 0$. Hence H_1 is 1-dimensional: one hole. ✓

Filled triangle. Same cycle, but now C_2 contains $[a, b, c]$, whose boundary is exactly that loop. So the cycle is a boundary, and $H_1 = 0$: no hole. ✓

And it is an invariant in the sense of §23: homeomorphic (indeed homotopy equivalent) spaces have isomorphic homology.

§25. From point clouds to barcodes

25.1 The problem with finite data

Homology is defined for shapes. But what we have on each side is a **finite set of points with distances between them** — a finite metric space. And the homology of a finite set of points is trivial: n isolated points, $b_0 = n$, everything else zero. Useless.

The fix: build a shape *from* the points, at a scale ϵ , and then don't commit to a value of ϵ .

25.2 The Vietoris-Rips complex

Definition. Given a finite metric space (X, d) and $\epsilon \geq 0$, the **Vietoris-Rips complex** $\text{VR}_\epsilon(X)$ is the simplicial complex whose k -simplices are the sets of $k + 1$ points that are pairwise within ϵ :

$$\{x_0, \dots, x_k\} \in \text{VR}_\epsilon(X) \iff d(x_i, x_j) \leq \epsilon \quad \forall i, j.$$

Note this depends on the points **only through the distance matrix**. No coordinates, no embedding, no basis. That is requirement (b) of §22, satisfied structurally: relabel or re-coordinatize however you like, the distances don't change, so the complex doesn't change.

Key monotonicity: if $\epsilon \leq \epsilon'$ then $\text{VR}_\epsilon(X) \subseteq \text{VR}_{\epsilon'}(X)$. Simplices only ever get added as ϵ grows.

25.3 Filtration and persistence

Definition. A **filtration** is the nested family $\{\text{VR}_\epsilon(X)\}_{\epsilon \geq 0}$, a growing sequence of complexes indexed by scale.

Since inclusion induces maps on homology, we get for each degree k a diagram of vector spaces and linear maps

$$H_k(\text{VR}_{\epsilon_1}) \rightarrow H_k(\text{VR}_{\epsilon_2}) \rightarrow H_k(\text{VR}_{\epsilon_3}) \rightarrow \dots$$

Definition. A **persistence module** is exactly such a family: a vector space V_ϵ for each ϵ and a linear map $V_\epsilon \rightarrow V_{\epsilon'}$ for each $\epsilon \leq \epsilon'$, composing consistently.

The point is that we can now track a hole *across scales*: when it appears (**birth**) and when it gets filled in (**death**).

25.4 The structure theorem and barcodes

IMPORTED (structure theorem for persistence modules). Under mild finiteness conditions, a persistence module decomposes uniquely as a direct sum of **interval modules** — pieces that are 1-dimensional on an interval $[b, d)$ and zero outside it.

Definition. The **barcode** of a filtration is the resulting multiset of intervals $\{[b_i, d_i)\}$. Equivalently, plotted as points (b_i, d_i) in the plane, the **persistence diagram**.

So the entire multi-scale homological content of a point cloud compresses to a list of intervals. Long bars are features that persist across many scales — robust structure. Short bars appear and vanish quickly — typically noise.

This is **persistent homology**, and it's the invariant $\mathcal{I}$ we were looking for.

25.5 Stability

IMPORTED (stability theorem, Cohen-Steiner-Edelsbrunner-Harer). If two finite metric spaces have distance matrices differing by at most δ (in sup norm), their persistence diagrams differ by at most δ in **bottleneck distance** — the cost of the best matching between the two diagrams, where the cost of matching two points is their ℓ^∞ distance and unmatched points are matched to the diagonal.

Wasserstein distance is the same idea with an ℓ^p sum over the matching rather than a max. It's the number Rung 1 computed.

Stability is what makes this usable on numerical data: finite- N noise degrades the barcode gracefully and boundedly, rather than catastrophically. A small perturbation of $G(\tau)$ cannot produce a wildly different barcode.

25.6 Why this satisfies §22

Requirement	How persistent homology meets it
(a) Same target on both sides	Both sides produce a barcode. Comparable by bottleneck/Wasserstein.
(b) Invariant under arbitrary choices	Depends only on the distance matrix. Coordinate-free by construction.
(c) Independent computation	Not automatic. It depends entirely on how the two distance matrices are built. This is where Rung 1 failed, and it's a discipline requirement, not a property of the tool.

§26. Categories, functors, and natural transformations

One more layer, and it's what turns the project from a numerical comparison into a theorem. This is the section that is *your* territory (Rung 5), so I'll build it carefully.

26.1 Categories

Definition. A **category** $\mathcal{C}$ consists of objects, and for each pair of objects a set of **morphisms** (arrows) between them, such that morphisms compose associatively and each object has an identity morphism.

Examples: **Vect** (vector spaces, linear maps); **Top** (topological spaces, continuous maps); **Simp** (simplicial complexes, simplicial maps). Also: the poset $(\mathbb{R}, \leq)$ is a category, with one arrow $\epsilon \rightarrow \epsilon'$ exactly when $\epsilon \leq \epsilon'$.

That last example is why persistence modules are categorical objects: **a persistence module is precisely a functor** $(\mathbb{R}, \leq) \rightarrow \mathbf{Vect}$.

26.2 Functors

Definition. A **functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object X of $\mathcal{C}$ an object $F(X)$ of $\mathcal{D}$, and to each morphism $f : X \rightarrow Y$ a morphism $F(f) : F(X) \rightarrow F(Y)$, preserving identities and composition: $F(g \circ f) = F(g) \circ F(f)$.

A functor is a **structure-preserving map between categories**. Not just objects to objects — the relationships come along too.

IMPORTED (standard). VR_ϵ and H_k are both functorial. Hence the whole pipeline "metric space $\rightarrow$ filtration $\rightarrow$ persistence module" is a functor.

This functoriality is what makes persistent homology well-behaved rather than an ad hoc recipe, and it's the property that lets us state the conjecture as a theorem.

26.3 Natural transformations

Definition. Given two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a **natural transformation** $\varphi : F \Rightarrow G$ assigns to each object X a morphism $\varphi_X : F(X) \rightarrow G(X)$, such that for every morphism $f : X \rightarrow Y$ in $\mathcal{C}$ the square

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow \varphi_X & & \downarrow \varphi_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

commutes: $\varphi_Y \circ F(f) = G(f) \circ \varphi_X$.

It is a **natural isomorphism** if every φ_X is an isomorphism.

The commuting square is the entire content, so read it in words: **the comparison φ is compatible with every morphism in the source category**. You don't just match objects one at a time — you match them in a way that respects all the structure relating them.

Contrast with a bare isomorphism. Two vector spaces of the same dimension are isomorphic, but you have to *choose* a basis; the isomorphism is not canonical, and it doesn't commute with anything in particular. A natural isomorphism is the canonical version: it works uniformly, for all objects at once, compatibly with all maps.

26.4 The conjecture, stated as a theorem

Now the payoff. Let $\mathcal{C}$ be a category whose objects are the physical setups (indexed by temperature, by scale, by whatever parameters both sides share) and whose morphisms are the natural maps between them. Define two functors into persistence modules:

$$F_{\text{SYK}} : \mathcal{C} \rightarrow \mathbf{Pers}, \quad F_{\text{JT}} : \mathcal{C} \rightarrow \mathbf{Pers}.$$

The categorical statement of ER=EPR (this project's central claim). There exists a natural transformation $\varphi : F_{\text{SYK}} \Rightarrow F_{\text{JT}}$, and it is a natural isomorphism.

What this buys, relative to "entanglement is geometry":

It is **checkable**: naturality is a commuting square you can verify or violate.

It is **falsifiable**: exhibit one square that fails to commute, or one φ_X that isn't invertible, and the claim is dead.

It is **diagnostic**: when it fails, it fails at a *specific object or morphism*, and that location tells you which structure broke.

It is **global**, not pointwise: you're not matching one number at one temperature, you're matching a coherent family across all of them simultaneously.

This is Rung 5, and it is yours: the categorical statement, the construction of φ , the proof of functoriality, and the proof of invertibility.

PART VI — THE LADDER

Five rungs. Each one: what we did, what it bought, what it did not buy.

§27. The design of the ladder

The strategy is to start with a case simple enough that every step can be checked by hand, then increase complexity one axis at a time:

Rung	System	New ingredient
1	Two entangled qubits	The pipeline itself
2	Thermofield double	Temperature as a parameter
3	SYK vs. JT at large N	Real physics on both sides
4	SYK at finite N	Quantum corrections
5	Categorical statement Theorem rather than computation	

§28. Rung 1 — two entangled qubits

What we did. Took a Bell-type state (§7.1), built a metric from its correlations, ran the full pipeline into a barcode (§25), constructed the gravity-side comparison object, and computed the Wasserstein distance between the two barcodes. Got zero.

What it bought. The pipeline runs end to end without exploding. Every stage — state to correlation matrix to distance matrix to filtration to barcode to distance — is implemented and produces output. That is real engineering value.

What it did not buy, and this is the most important lesson in the project.

It was **tautological**. Two separate mechanisms made it so:

φ was defined as the ratio of two quantities that had been *defined to be equal*.

Ryu–Takayanagi (§15) was used as an **input** to the gravity side, not as something the gravity side independently produced.

So Wasserstein = 0 was algebraically guaranteed before a single number was computed. Apply the §22 test: *what input would have made these disagree?* Answer: none. There was no possible world in which that number came out nonzero.

The principle, stated permanently. Agreement is evidence only if the agreeing things are independent. Construction-guaranteed agreement must be identified and quarantined *before* it can be counted as a result. Pipeline validation is not a physics result.

This is the same discipline as the Claude Code review protocol: an assertion that couldn't have come out otherwise isn't a verification. It's a restatement.

§29. Rung 2 — temperature dependence

What we did. Swept β across the thermofield double family (§12). Watched entanglement entropy run from maximal at $\beta \rightarrow 0$ down to zero at $\beta \rightarrow \infty$, exactly as §12.3 predicts. Tracked the corresponding horizon/wormhole quantity across the same sweep. Monotone, smooth, no pathologies anywhere.

What it bought. Pipeline validation over a one-parameter *family* rather than a single point. Genuinely more useful than Rung 1 — a bug that happens to give the right answer at one point rarely survives a whole sweep.

What it did not buy. The same tautology, now parameterized. The gravity-side number was still the entanglement entropy pushed through RT, so it was the entanglement wearing a hat. A monotone curve matching a monotone curve, when one was computed from the other, is not two independent computations agreeing.

§30. Rung 3 — SYK vs. JT, and the structural mismatch

This is where the project became real, because it broke in an informative way.

30.1 The mismatch

The first construction indexed the two point clouds like this:

SYK side: by fermion flavour $i = 1, \dots, N$, from the correlation matrix.

JT side: by radial position in the bulk geometry.

These index sets are **not related by the holographic dictionary**. There is no map sending fermion flavour i to a bulk radius. Recall §18.1: the SYK index is a label with no geometric meaning whatsoever — the model has no spatial structure by design.

So the comparison wasn't giving the wrong answer. It was **ill-posed**. The two barcodes were barcodes of incomparable things.

Structural lesson. Before comparing two constructions, check that they are indexed by the same thing. This sounds obvious written down. It was not obvious in the code.

30.2 The resolution: index both sides by τ

Both sides get indexed by **Euclidean time on the thermal circle** (§11.3).

SYK side: points are times τ_a on the circle of circumference β ; the distance between τ_a and τ_b is built from the two-point function $G(\tau_a - \tau_b)$ (§19.1).

JT side: points are boundary times; the distance is the regularized geodesic length between the corresponding boundary points (§13).

Now both functors take the same input category, and the comparison of §26.4 is well-formed. This is why §11 was flagged as the most important piece of technology in the document: the thermal circle is what makes the two sides commensurable at all.

(a) 1D point clouds have trivial H_1 at every scale.

PROVEN, via chordal indifference graphs. (Note: the curvature and triangle-filling arguments we tried first were *wrong*. The chordality argument is the correct one.)

Consequence: any naive one-dimensional embedding produces an empty barcode in degree 1. Nothing to compare. **You need the circle** — which, by §11.3, is what the physics hands you anyway.

(b) Adamaszek–Adams: the persistence module collapses to two scalars.

IMPORTED (Adamaszek–Adams). The homotopy types of Vietoris–Rips complexes of evenly spaced points on a circle are completely known.

Applied to a **monotone circulant thermal-circle filtration**, this means the entire persistence module is determined by just **two Fourier modes** of the correlator, $\hat{G}(1)$ and $\hat{G}(8)$, weighted approximately $23 : 1$.

This is the crown jewel of the project so far. An apparently infinite-dimensional barcode comparison reduces, exactly, to two numbers. It is a genuine structure theorem, it is yours, and it is what the paper should be built around.

(c) UV localization. The barcode is dominated by short- τ structure in G — the singular behaviour as $\tau \rightarrow 0$ noted in §19.4. The deep-IR Schwarzian sector, which is *precisely* the part that is supposed to match gravity (§21), barely registers in the barcode.

That mechanism explains everything Rung 4 found.

§31. Rung 4 — finite N , and an honest negative

31.1 The plan

Measure how far the SYK barcode sits from the Schwarzian/JT prediction, and test whether the deviation scales like $1/N$ — as genuine quantum corrections should.

31.2 What happened

Committed at `3ee3035` and `a2204e5` with closure text, verbatim rescore output, and the proven-vs-conjectured ledger in `rung4_results.txt`.

C_{fit} is β -**dependent**. Spreads of 29–158% across β at fixed N . A genuine dictionary constant should not drift with temperature (§19.5).

The power law $C_{\text{fit}} = aN^b$ gives b with large uncertainty. No clean $1/N$ signal to extract.

**** $C_{\text{fit}}(\hat{G}_1)/C_{\text{fit}}(\hat{G}_8)$ ranges 1.47–7.81.**** The two surviving scalars from §30.3(b) want *different* calibrations, so no single overall normalization reconciles them.

The result that survives everything: shape failure of the Schwarzian profile at accessible N , established independent of calibration.

That last point is the real content. It is not a normalization problem you could fix with a better constant — the profile has the wrong *shape*. Calibration-independence is what makes it a result rather than a fitting complaint.

The Maldacena–Stanford large- N dictionary is declared explicitly as an **imported calibration**, not a measured output. No repeat of the Rung 1 sin.

31.3 A correction worth re-reading

The claim “no C can reach SYK” was originally overgeneralized from a single- β sweep, and the $\beta = 40$ row’s interior minimum refuted it.

Scoping discipline: reachability claims require full β coverage. The final claim is scoped to what the sweep actually supports, and no further.

31.4 Also closed

The Jacobian question (`LOG_JACOBIAN = -log(2)`, `schwarzian.py` line 68) was resolved by A/B test. Closed; not to be reopened.

§32. Where this leaves the project

The negative isn’t a failure. It’s a **structure theorem with a falsifiable criterion attached**, which is a strictly better paper than the one originally planned.

Reframed structure:

Structure theorem on circulant thermal-circle filtrations: the persistence module reduces to two scalars (§30.3b).

Corollary: exactly when φ is a natural isomorphism (§26.4).

Application: the SYK/JT comparison, where the criterion demonstrably fails at $N \leq 18$ (§31).

Methodological conclusion: two-boundary constructions are the correct next step.

And here’s the physical punchline that makes point 4 sing:

A single thermal circle has no wormhole in it.

The thermofield double is a *two-sided* object (§12, §16). If the Einstein–Rosen bridge is going to appear as a topological feature of a barcode, the filtration has to see **both boundaries at once**. A one-boundary construction cannot see it, and combined with the UV-localization mechanism of §30.3(c), that’s not a hypothesis — it’s the explanation for why Rung 4 came out the way it did.

So Rung 4 didn't fail to find the wormhole. It established that a one-boundary construction *cannot* find it. That's a theorem, not a null result, and it is the natural on-ramp to Rung 5.

Rung 5, which is yours: construct φ in the two-boundary category, where it has an actual chance of being invertible, and prove functoriality and invertibility via the structure theorem.

Dependency ledger — everything imported, in one place

For the paper. Consistent treatment of everything we use but did not derive:

Result	Source	Where used
Born rule	Axiom of QM	§4.2
Bekenstein–Hawking entropy	Bekenstein, Hawking	§14
Hawking temperature	Hawking	§14
Chaos bound	Maldacena–Shenker–Stanford	§14, §18.2
AdS/CFT	Maldacena	§15
Ryu–Takayanagi	Ryu–Takayanagi	§15 — and the source of the Rung 1 tautology
TFD $\leftrightarrow$ eternal black hole	Maldacena	§16
Melonic dominance	Large- N diagram counting	§19.3
Schwarzian from SYK; value of C	Maldacena–Stanford	§19.5, §31
JT boundary dynamics = Schwarzian	Standard JT	§20
Schwarzian correlator	MTV 1705.08408	Rung 4 fits
Persistence structure theorem	Standard (Zomorodian–Carlsson etc.)	§25.4
Stability theorem	Cohen–Steiner–Edelsbrunner–Harer	§25.5
Functoriality of VR and H_k	Standard	§26.2
VR homotopy types on the circle	Adamaszek–Adams	§30.3b

Derived here and reproducible cold: §4.2 expectation values, §5.3 stationarity, §7.1 Bell entanglement, §8.2–8.4 density matrix and operational content, §8.5 Bell reduced state, §9.2 Schmidt, §10.2 Gibbs from maximum entropy, §11 Euclidean–thermal identification, §12.2 the TFD theorem, §24.2–24.3 homology, §30.3a trivial H_1 in 1D.

End. Read §11, §22, §26, and §32 twice — those four carry the argument.